Zhihao Chen

MSc Data Science Student and AI Researcher

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Education

MSc **Federal Institute of Technology Lausanne (EPFL)**, Data Science — Lausanne, Switzerland, Sept 2026 - present

BSc **Polytechnic University of Catalonia (UPC)**, Artificial Intelligence — Barcelona, Spain, Sept 2022 - June 2026
- GPA: 9.00/10.0 (Top 10%)

Exchange **Federal Institute of Technology Lausanne (EPFL)**, Computer Science — Lausanne, Switzerland, Sept 2025 - Feb 2026
- GPA: 5.93/6.0

Experience

VILAB - EPFL, Graduate Student Researcher — Lausanne, Switzerland, Sept 2026 - present, 3 months
Continuous Generative Modeling for Unified Any-to-Any Multimodal Systems
- Extending unified any-to-any multimodal architectures via continuous generative processes and flow-based decoders across heterogeneous sensory modalities.

EPFL, Student Teaching Assistant — Lausanne, Switzerland, Sept 2026 - present, 2 months
EE-556 Mathematics of Data: From Theory to Computation

LAMP - CVC - UAB, Undergraduate Student Researcher — Barcelona, Spain, Feb 2026 - Sept 2026, 8 months
Unified Multi-Modal Continual Learning
- Formulated a sequential benchmark to measure cross-modal catastrophic forgetting between visual reasoning and image synthesis in unified multimodal generative models.
- Found that decoupled architectures mitigate bidirectional representation degradation compared with monolithic architectures.
- Awarded the Elena Maseras Research Scholarship 2026 for academic excellence (13% acceptance rate).

IRI - UPC, Undergraduate Student Researcher — Barcelona, Spain, Feb 2026 - Sept 2026, 8 months
Future Action Prediction in Video Models
- Developed WarmFlow, an encoder-initialized flow-matching framework that replaces pure-noise diffusion with dynamic structural priors for action segmentation.
- Achieved competitive generative coherence with one function evaluation (NFE=1), reducing computation by 10x-25x for latency-critical applications.

BioRob - EPFL, Visiting Student Researcher — Lausanne, Switzerland, Sept 2025 - Feb 2026, 6 months
Context-Aware Spatial Action Recognition
- Extended the SkateFormer architecture for multimodal spatial action recognition, using geometric priors to generalize to occluded layouts.
- Curated a 10k-sample dataset and deployed the framework on a ROS2 mobile swarm, achieving a 20% accuracy gain over baselines and 25ms latency for closed-loop control.

PayRetailers, AI Engineer Intern — Barcelona, Spain, June 2025 - Aug 2025, 3 months
Multi-Agent Systems and LLM Analysis
- Evaluated prompting strategies, including chain-of-thought and few-shot prompting, and connected LLMs to external tools through multi-agent workflows.
- Built internal B2B AI solutions, including retrieval-augmented generation systems and automated analytical pipelines.

Publications

Continual Adaptation Across Modalities: Learning and Forgetting in Unified Multimodal Models
Xinrui Wang, **Zhihao Chen**, Jia Tang, Alexandra Gomez-Villa, Bartłomiej Twardowski, Joost van de Weijer, Songcan Chen
arxiv.org/abs/2610.38436 (NeurIPS 2026 Workshop on CL4FMAgents)

Projects

B2Travel - HackUPC 2026
Built an LLM-grounded multimodal recommendation system that maps CLIP visual-semantic embeddings into a 3D topological space for spatial retrieval.
- Second Prize, Skyscanner Challenge; Best Use Award of ElevenLabs.

MXEgg - LauzHack 2025
Built an LLM-driven multi-agent simulation with a custom physics engine and localized vision to study emergent cooperative behavior.
- Overall finalist, LauzHack 2025.

AIvidence - MERIThon 2025 AI Challenge
Built a retrieval-grounded multi-agent framework that distributes chain-of-thought reasoning across specialized LLMs for deepfake detection.
- First Prize, MERIThon 2025 AI Challenge.

Awards

Elena Maseras Research Scholarship 2026 — LAMP - CVC - UAB
Awarded by LAMP, CVC-UAB for academic excellence (13% acceptance rate).

Santander Excellence Scholarship — Santander 360º
An award recognizing top academic performance as well as participation in cultural, social, and professional development activities (only 77 recipients selected across all Spanish universities and academic fields).

Second Prize, Skyscanner Challenge & Best Use Award of ElevenLabs — HackUPC 2026

Overall Finalist, LauzHack 2025 — EPFL, Switzerland

First Prize, MERIThon 2025 AI Challenge — UPC, Spain

Academic Honors — UPC, Spain
Four courses with distinction.

Skills

Programming: Python, C, C++, MATLAB, SQL, R

Machine Learning and Data: PyTorch, Scikit-learn, Hugging Face, OpenCV, LangChain, Weights & Biases, DDP/FSDP

Systems and Engineering: Jupyter, LaTeX, Git/GitHub, Docker, SLURM

Languages

Chinese: Native proficiency

Spanish: Native proficiency

Catalan: Native proficiency

English: Professional proficiency (C1)

French: Elementary proficiency (A2)